Insight46: Comparison between Freesurfer and GIF for amyloid PET quantification

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Document details

Categories of variables	Longitudinal detailed regional (Desikan-Killiany parcellations) amyloid-beta PET [18F]florbetapir SUVRs and amyloid-beta status for phase 1 and 2 Insight 46 scans using FreeSurfer, MRI-based anatomical parcellations.
Summary of work undertaken	This is an additional spreadsheet to the original FreeSurfer based amyloid PET measures that provides the SUVR values for the individual Desikan-Killiany (DK) atlas regions. We do not provide positive or negative amyloid status for each DK region. This is only provided at a composite and lobar regions in the previous spreadsheet. See below for details of methodology. Imran converted dataset to STATA and have it research ready for Condor
Names of source variables	Each, source, file, contains, the, following, variables:, subject, session, assigned', Cerebral, White, Matter', Lateral, Ventricle', Inf, Lat, Vent', Cerebellum, White, Matter', Cerebellum, Cortex', Thalamus, Proper', Caudate', Putamen', Pallidum', Ventricle', Ventricle', Stem', Hippocampus', Amygdala', Accumbens, area', VentralDC', vessel', choroid, plexus', Cerebral, White, Matter', Lateral, Ventricle', Inf, Lat, Vent', Cerebellum, White, Matter', Cerebellum, Cortex', Thalamus, Proper', Caudate', Putamen', Pallidum', Hippocampus', Amygdala', Accumbens, area', VentralDC', vessel', choroid, plexus', hypointensities', Chiasm', frontal, parietal, occipital, temporal, cingulate, insula, composite, hippocampus, caudate, putamen, thalamus, cortical_gm, subcortical_gm, WM, hypointensities', Ventricle', exclude
Output variables	I46P2_AmyloidPET3
Papers using these variables	N/A

Method

** **Please see pdf from previous dataset: fs_vs_gif_validate.pdf

Purpose

This document describes how the Insight46 phase 2 amyloid PET regional SUVR file is read into Stata and cleaned. The do-file reads the CSV, renames and labels the SUVR variables, and codes missing data. Measurement values are not altered.

1. The input file

1.1 Layout

- One row per participant record; 165 columns.
- Five identifying and descriptive columns: subject, comment_bl, comment_fu, session_date_bl, session_date_fu.
- 160 SUVR columns: 80 regions, each with a baseline (*bl) and a follow-up (_fu) value, using the whole cerebellum as the reference region (_wc*).
- The data dictionary (..._DATADICTIONARY.csv) gives the raw variable name and description of each SUVR column.

1.2 Missing data in the raw file

- "NA" is written where there is no SUVR value.

- Because of the "NA" entries, Stata treats the SUVR columns as text.

1.3 Points to check in each extract

- The SUVR columns contain no blank cells in that extract, only "NA".

2. Why a standard import fails

Stata variable names can have at most 32 characters. 24 of the raw column names are longer, for example suvr_lh_rostralanteriorcingulate_wc_bl (38 characters). When cut to 32 characters the baseline and follow-up names become identical, so a normal import delimited renames them v# and any code that refers to the raw names fails. The do-file avoids the problem by reading every column as text with no names (v1 to v165, header in row 1) and then naming the columns by their position.

3. Steps in the do-file

3.1 Import

Set the path on the local infile line, then read every column as text without variable names. import delimited using "`infile'", varnames(nonames) stringcols(_all) encoding("utf-8") clear

3.2 Check the header and remove it

Row 1 holds the column names. The do-file checks the number of columns and the first, sixth and last names, then drops row 1. If the file layout changes, the check stops the run instead of mislabelling columns. assert c(k) == 165 assert v1 == "subject" in 1 assert v6 == "suvr_left_amygdala_wc_bl" in 1 assert v165 == "suvr_rh_insula_wc_fu" in 1 drop in 1

3.3 Name the non-SUVR columns

v1 to v5 are renamed subject, comment_bl, comment_fu, session_date_bl and session_date_fu. They stay as text. Dates need converting once they are filled in. They are dropped from the final dataset.

3.4 Value label for no data

The value label nodata_i46p2 is created with -9 labelled "No data" only if it does not already exist. capture label list nodata_i46p2 if _rc != 0 label define nodata_i46p2 -99 "No data" -9 “Unknown” -7 2 “No Insight phase 2 data”

3.5 Convert, recode and label each SUVR variable

Each of the 160 columns has the same four commands. real() turns text into a number and gives a missing value for text such as "NA". "NA" is then recoded to -9, and the value label and variable label are applied. generate double svl_amg_wc_bl_i46p2 = real(v6) replace svl_amg_wc_bl_i46p2 = -9 if v6 == "NA" label values svl_amg_wc_bl_i46p2 nodata_i46p2 label variable svl_amg_wc_bl_i46p2 "..."

3.6 Check for unexpected text

A loop reports any value that is not a number, "NA" or blank, and ends with the number of variables affected. Zero means the conversion was clean.

3.7 Optional: code blanks as -7

To code originally blank values as -7, run this after the conversion blocks. At that point "NA" has already become -9, so the only missing values left are the originally blank ones. foreach v of local newlist { replace `v' = -7 if missing(`v') } label define nodata_i46p2 -7 "Blank", modify Each command must be on its own line. If

pasted onto a single line, everything after a * is treated as a comment and nothing runs.

3.8 Remove the temporary columns

Once the check is clean the temporary text columns are dropped (drop v6-v165), the data are compressed, and the file can be saved.

4. Names and labels

4.1 Variable names

New names are built as: sv + hemisphere letter (l or r) + _ + region abbreviation + *wc* + bl or fu + _i46p2.

Raw variable	New variable
suvr_left_amygdala_wc_bl	svl_amg_wc_bl_i46p2
suvr_left_amygdala_wc_fu	svl_amg_wc_fu_i46p2
suvr_rh_insula_wc_bl	svr_ins_wc_bl_i46p2

Left (left, lh) becomes l and right (right, rh) becomes r. All new names are unique and no longer than 20 characters.

4.2 Region abbreviations

Region	Abbreviation
amygdala	amg
hippocampus	hip
thalamus proper	tha
caudate	cau
putamen	put
pallidum	pal
banks of the superior temporal sulcus	bsts
caudal anterior cingulate	cacg
caudal middle frontal	cmf
cuneus	cun
entorhinal	ent
fusiform	fus
inferior parietal	ip
inferior temporal	it
isthmus cingulate	icg
lateral occipital	lo
lateral orbitofrontal	lof
lingual	lin
medial orbitofrontal	mof
middle temporal	mt
parahippocampal	phip
paracentral	pcen
pars opercularis	pop

pars orbitalis	por
pars triangularis	ptr
pericalcarine	pcal
postcentral	poc
posterior cingulate	pcg
precentral	prc
precuneus	pcun
rostral anterior cingulate	racg
rostral middle frontal	rmf
superior frontal	sf
superior parietal	sp
superior temporal	st
supramarginal	sm
frontal pole	fp
temporal pole	tp
transverse temporal	tt
insula	ins

4.3 Variable labels

Each label is the data dictionary description without the leading "baseline" or "follow-up" and with a capital first letter, followed by the study and time point.

Baseline

- Longitudinal (phase 1 and 2 only) - baseline - Insight46 phase 1

Follow-up

- Longitudinal (phase 1 and 2 only) - follow-up - Insight46 phase 2

Label length

Stata limits variable labels to 80 characters, and these labels are longer (up to 169 characters). After running, check one label with describe to see whether it has been cut.

5. Missing value codes

Code	Meaning	Source
-9	No data	"NA" in the raw file
-7	Blank	Empty cell in the raw file (optional step 4.7)

Both codes use the value label nodata_i46p2. Treat them as missing in analyses, for example with mvdecode or by excluding them in the model. **7. Supporting files**

File	Contents
read_and_rename_suvr_i46p2.do	Imports, renames, labels and recodes the data.

suvr_rename_labels_map.xlsx	One row per variable: raw name, old label (data dictionary), new name and new label.

Variables in this document

160 high-confidence matches

This list is very large (160 variables) and is not reproduced in this PDF. Please visit the OWL website to see the full list.

3 low-confidence matches

Variable	Label	How it was found
Population characteristics [54] › Baseline characteristics [401]		
inf	Inflation factor to adjust for stratified sample	prose scan
Questionnaire data [55] › Health and medical history [510] › Physical capability [5108]		
step	Do you find it difficult walking up and down stairs due to long term health problems? - at age 53 years	prose scan
Questionnaire data [55] › Sociodemographics [515] › Household [5152]		
own	Now I would like to get some general information about your household. Does your household own or rent this accommodation? - at age 53 years	prose scan